Civilization λ Attractor Design: Lyapunov Function and Minimal Institutional Core

SCX Equality Theory Framework · C6 Extension

Lyapunov Stability, Basin of $\lambda > 0$ Attraction, Institutional Phase Space

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Abstract

How do civilizations maintain long-term stability? The C5 extension (Civilization Inequality Gauge) of the SCX Equality Theory framework explained the paradox of long-lived unequal civilizations through coupling suppression κ. C6 pushes deeper: **does there exist a minimal institutional core such that civilization dynamics necessarily converge to an attractor with positive Lyapunov exponent $\lambda > 0$?** This paper introduces the **Lyapunov civilization function** $V(\mathbf{x})$ mapping civilization state $\mathbf{x} \in \mathcal{X}$ (institutional configuration space) to a stability measure. Core contributions: (1) establishing the civilization SDE model $\frac{d}{dt}\mathbf{x}_t = -\nabla V(\mathbf{x}_t) + \sqrt{2D}\, d\mathbf{W}_t$, proving that when V is radially unbounded with compact level sets, the system converges to the $\lambda > 0$ basin with probability 1; (2) defining the **Minimal Institutional Core** (MIC) as the minimal institutional configuration set guaranteeing $\lambda > 0$, and proving its existence; (3) proving the MIC dimension lower bound of 3 (corresponding to the attractor versions of C5's triple levers: information isolation, force monopoly, belief legitimacy); (4) establishing the exponential relationship $T \propto \exp(\lambda \cdot \tau_{\text{char}})$ between λ and civilization lifespan T; (5) numerical verification: demonstrating attractor basin structure, critical phase transitions, and MIC reconstruction in 2-institution through N-institution models. The paper is rigorously formalized with a complete verification script.

Keywords: Lyapunov function, civilization attractor, λ stability index, minimal institutional core, basin of attraction, SDE, phase transition, SCX framework, C6 extension.

Contents

1 Introduction: From C5 Coupling Suppression to C6 Attractor Design

1.1 C5 Recap and C6 Motivation

The core finding of C5 (Civilization Inequality Gauge) is: civilization lifespan T depends not directly on inequality Δ, but on the effective disruption rate $\kappa_{\text{eff}} \cdot \Delta^2$, where $\kappa_{\text{eff}} \in [0, 1]$ is the coupling strength through which subordinates perceive elite states. Through the triple mechanisms of information isolation, force monopoly, and belief legitimacy, κ_{eff} can be suppressed arbitrarily close to zero, granting extremely long lifespans to highly unequal civilizations.

However, C5 left a fundamental open question: **suppression itself consumes institutional resources. Does the institutional configuration needed to sustain suppression possess a stable attractor structure? In other words, does there exist a set of institutional features such that civilization dynamics, even under perturbation, automatically return to a $\lambda > 0$ (stable) state?**

C6 answers this question by introducing Lyapunov function and attractor basin theory. We model civilization as a stochastic dynamical system in institutional phase space, whose stability is measured by the Lyapunov exponent λ. $\lambda > 0$ indicates the civilization is within a stable attractor basin — local perturbations decay and the system returns to equilibrium; $\lambda < 0$ indicates the civilization is in a repeller region — small perturbations are amplified and the system heads toward collapse.

1.2 Core Questions and Contributions

This paper addresses six core questions:

Q1 Existence of Lyapunov function. Does there exist a scalar function $V(\mathbf{x})$ on civilization institutional space whose gradient flow describes long-term civilizational evolution?

Q2 $\lambda > 0$ basin structure. What is the geometric and topological structure of the $\lambda > 0$ attractor basin? How do basin boundaries vary with institutional parameters?

Q3 Minimal Institutional Core (MIC). What is the minimal institutional dimensionality needed to enter and remain in the $\lambda > 0$ basin? What are its mathematical characteristics?

Q4 Noise robustness. How do stochastic perturbations (external invasion, natural disasters, technological shocks) affect attractor basin stability? Is there a critical noise threshold?

Q5 λ–T relationship. What is the quantitative relationship between Lyapunov exponent λ and expected civilization lifespan T?

Q6 Phase transitions and critical phenomena. In institutional parameter space, does there exist a critical surface across which the system abruptly transitions from $\lambda > 0$ to $\lambda < 0$?

2 Formal Theory of the Lyapunov Civilization Function

2.1 Institutional Phase Space

Definition 2.1 (Institutional Phase Space)**.** *The state of civilization $\mathcal{C}$ is characterized by an institutional configuration vector $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$, where $d \geq 3$ is the institutional dimensionality. $\mathcal{X}$ is a compact or locally compact institutional phase space whose coordinate axes include but are not limited to:*

- x_1*: Information isolation intensity (*0 *= fully transparent,* 1 *= fully isolated)*
- x_2*: Force concentration (*0 *= fully dispersed,* 1 *= full monopoly)*
- x_3*: Belief legitimacy (*0 *= fully secularized,* 1 *= fully sacralized)*
- x_4*: Resource redistribution rate*
- x_5*: Elite rotation rate*
- $\cdots$

The institutional phase space is equipped with a Riemannian metric $g_{ij}(\mathbf{x})$ encoding the switching costs between different institutional dimensions.

Definition (Institutional Phase Space). The state of civilization $\mathcal{C}$ is characterized by an institutional configuration vector $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$, where $d \geq 3$ is the institutional dimensionality. $\mathcal{X}$ is a compact or locally compact institutional phase space whose coordinate axes include but are not limited to the above dimensions. The phase space is equipped with a Riemannian metric $g_{ij}(\mathbf{x})$ encoding the switching costs between different institutional dimensions.

2.2 Lyapunov Function Definition

Definition 2.2 (Lyapunov Civilization Function)**.** *The* ***Lyapunov civilization function*** *$V : \mathcal{X} \to \mathbb{R}_{\geq 0}$ is a C^2 function satisfying the following conditions:*

(i) ***Positive definiteness:*** *$V(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathcal{X}$, and $V(\mathbf{x}) = 0$ if and only if $\mathbf{x} \in \mathcal{A}^*$ (global optimal attractor set).*

(ii) ***Radial unboundedness:*** *$\lim_{\|\mathbf{x}\| \to \infty} V(\mathbf{x}) = \infty$ (if $\mathcal{X}$ is non-compact).*

(iii) ***Dissipativity:*** *Along the noiseless dynamics $\dot{\mathbf{x}} = -\nabla V(\mathbf{x})$, we have $\frac{d}{dt}V(\mathbf{x}_t) = -\|\nabla V(\mathbf{x}_t)\|^2 \leq 0$.*

(iv) ***Institutional interpretability:*** *$V(\mathbf{x})$ decomposes into contributions from institutional dimensions:*

$$V(\mathbf{x}) = \sum_{k=1}^{d} w_k \cdot \ell_k(x_k) + \sum_{i<j} J_{ij} \cdot \phi_{ij}(x_i, x_j), \tag{1}$$

where ℓ_k is the single-dimension institutional tension, ϕ_{ij} is the inter-dimension interaction energy, and $w_k > 0$, $J_{ij} \in \mathbb{R}$ are coupling weights.

Physically, $V(\mathbf{x})$ measures the distance between the current institutional configuration and the optimal stable configuration — larger V implies greater instability and a stronger tendency to evolve toward lower-V regions.

Definition (Lyapunov Civilization Function). The **Lyapunov civilization function** $V : \mathcal{X} \to \mathbb{R}_{\geq 0}$ is a C^2 function satisfying the above conditions. Physically, V measures the distance between the current institutional configuration and the optimal stable configuration — larger V implies greater instability and a stronger tendency to evolve toward lower-V regions.

Remark 2.3 (Difference from Physical Lyapunov Functions)**.** *In classical mechanics, Lyapunov functions decrease to zero along trajectories, corresponding to stability. Here we invert the sign convention: $V(\mathbf{x})$ decreases along gradient flow, but the **Lyapunov exponent** λ (defined below) is positive near local minima of V — $\lambda > 0$ corresponds to a stable attractor. This aligns λ's sign directly with civilizational health: $\lambda > 0$ healthy, $\lambda < 0$ pathological.*

2.3 Definition of Lyapunov Exponent

Definition 2.4 (Civilization Lyapunov Exponent)**.** *Let $\mathbf{x}^*$ be a local minimum of V (i.e., an institutional equilibrium). Linearizing the dynamics $\dot{\mathbf{x}} = -\nabla V(\mathbf{x})$ near this point yields the Jacobian matrix:*

$$\mathbf{J}(\mathbf{x}^*) = -\nabla^2 V(\mathbf{x}^*), \tag{2}$$

*where $\nabla^2 V$ is the Hessian of V. The **civilization Lyapunov exponent** λ is defined as the maximum real part of the eigenvalues of $\mathbf{J}(\mathbf{x}^*)$:*

$$\lambda(\mathbf{x}^*) = \max_i \Re(\mu_i), \quad \text{where } \mu_i \text{ are the eigenvalues of } \mathbf{J}(\mathbf{x}^*). \tag{3}$$

Since $\nabla^2 V(\mathbf{x}^*)$ is positive (semi-)definite at a local minimum, the eigenvalues of $\mathbf{J} = -\nabla^2 V$ are all non-positive. Therefore $\lambda(\mathbf{x}^*) \leq 0$, and:

- *$\lambda < 0$: $\mathbf{x}^*$ is an **asymptotically stable** attractor (center of the stable basin);*
- *$\lambda = 0$: $\mathbf{x}^*$ is **critically neutral** (boundary case);*
- *$\lambda > 0$: impossible at a Lyapunov minimum (since eigenvalues of $\mathbf{J}$ cannot be positive).*

Note on sign convention: *Throughout this paper, λ refers to the stability index: $\lambda \equiv -\max_i \Re(\mu_i)$, where μ_i are the eigenvalues of $-\nabla^2 V$. Under this convention, $\lambda > 0$ = stable attractor, $\lambda < 0$ = unstable (saddle or repeller), $\lambda = 0$ = neutral. Equivalently, $\lambda \equiv \lambda_{\min}(\nabla^2 V(\mathbf{x}^*))$, the smallest eigenvalue of the Hessian.*

Definition (Civilization Lyapunov Exponent). Let $\mathbf{x}^*$ be a local minimum of V. The **civilization Lyapunov exponent** λ is defined as above. **Note on sign convention:** Throughout this paper, λ refers to the stability index: $\lambda \equiv \lambda_{\min}(\nabla^2 V(\mathbf{x}^*)) > 0$, i.e., the smallest eigenvalue of the Hessian. Under this convention, $\lambda > 0$ = stable attractor, $\lambda < 0$ = unstable (saddle or repeller), $\lambda = 0$ = neutral.

2.4 Gradient Flow and SDE Description

Definition 2.5 (Civilization Dynamics SDE)**.** *The stochastic evolution of the institutional configuration* $\mathbf{x}_t$ *is governed by the following Itô SDE:*

$$d\mathbf{x}_t = -\nabla V(\mathbf{x}_t)\,dt + \sqrt{2D(\mathbf{x}_t)}\,d\mathbf{W}_t, \tag{4}$$

where:

- $-\nabla V$ *is the deterministic drift term, driving the system toward lower-*V *(more stable) states;*
- $\mathbf{W}_t$ *is a d-dimensional standard Brownian motion, representing institutional noise (external shocks, internal fluctuations);*
- $D(\mathbf{x}) \geq 0$ *is the state-dependent diffusion coefficient (noise intensity). Common choices include:*
  - *Constant diffusion:* $D(\mathbf{x}) \equiv D_0$ *(uniform noise)*
  - *Institution-dependent diffusion:* $D(\mathbf{x}) = D_0 \cdot (1 + \alpha \|\mathbf{x} - \mathbf{x}^*\|^2)$ *(noise increases away from equilibrium)*

Definition (Civilization Dynamics SDE). The stochastic evolution of institutional configuration $\mathbf{x}_t$ is governed by the Itô SDE above.

Proposition 2.6 (Stationary Distribution)**.** *If* V *is radially unbounded and* $D(\mathbf{x}) > 0$ *everywhere, then SDE (**??**) has a unique stationary distribution (Gibbs measure):*

$$\pi(\mathbf{x}) = \frac{1}{Z}\exp\left(-\frac{V(\mathbf{x})}{D_0}\right), \quad Z = \int_{\mathcal{X}} \exp\left(-\frac{V(\mathbf{x})}{D_0}\right) d\mathbf{x}, \tag{5}$$

where D_0 *is the uniform diffusion constant and* Z *is the partition function. The stationary distribution concentrates probability mass near the global minimum of* V *— the civilization necessarily resides near a stable attractor in the long-term limit.*

Proof. Stationary solution of the Fokker–Planck equation. When D is constant, $\frac{\partial \pi}{\partial t} = \nabla \cdot (\pi \nabla V) + D_0 \nabla^2 \pi = 0$ admits the general solution $\pi \propto \exp(-V/D_0)$. Radial unboundedness guarantees $Z < \infty$ (normalizability). Uniqueness follows from elliptic operator theory. □

3 Theory of the $\lambda > 0$ Attractor Basin

3.1 Basin Definition and Geometry

Definition 3.1 ($\lambda > 0$ Attractor Basin)**.** *Let* $\mathcal{A}^* = \{\mathbf{x}^* \in \mathcal{X} : V(\mathbf{x}^*) = \min_{\mathcal{X}} V\}$ *be the set of global Lyapunov minima. For each* $\mathbf{x}^* \in \mathcal{A}^*$*, define its* ***basin of attraction****:*

$$\mathcal{B}(\mathbf{x}^*) = \left\{\mathbf{x}_0 \in \mathcal{X} : \lim_{t\to\infty} \mathbf{x}_t = \mathbf{x}^* \text{ a.s. along deterministic flow } \dot{\mathbf{x}} = -\nabla V(\mathbf{x})\right\}. \tag{6}$$

The $\lambda > 0$ ***basin*** $\mathcal{B}_+ \subseteq \mathcal{X}$ *is the union of all states satisfying* $\lambda(\mathbf{x}) > 0$ *(under the stability index convention):*

$$\mathcal{B}_+ = \left\{\mathbf{x} \in \mathcal{X} : \lambda_{\min}(\nabla^2 V(\mathbf{x})) > 0\right\}. \tag{7}$$

The ***basin boundary*** $\partial\mathcal{B}_+$ *is the* $\lambda = 0$ *hypersurface where the system undergoes a stability phase transition.*

Definition ($\lambda > 0$ Attractor Basin). The basin of attraction and $\lambda > 0$ basin are defined as above. The basin boundary $\partial\mathcal{B}_+$ is the $\lambda = 0$ hypersurface where the system undergoes a stability phase transition.

Proposition 3.2 (Basin Volume Lower Bound). *Suppose V has K local minima on $\mathcal{X}$, corresponding to attractors $\mathbf{x}_1^*, \ldots, \mathbf{x}_K^*$. Let r_k be the shortest distance (under metric g_{ij}) from the k-th attractor to its basin boundary. Then the basin volume satisfies:*

$$\mathrm{Vol}(\mathcal{B}_+) \geq \sum_{k=1}^{K} \omega_d \cdot r_k^d, \tag{8}$$

where $\omega_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the d-dimensional unit ball.

Key implication: ***The higher the institutional dimension d, the more sensitive the basin volume is to the attractor radius*** *($\propto r_k^d$), so high-dimensional civilizations are more prone to losing stability through institutional degradation (shrinking r_k).*

Proof. Each attractor $\mathbf{x}_k^*$ has a basin containing at least a geodesic ball of radius r_k (local strict convexity is guaranteed by the positive-definite Hessian). The ball volume is $\omega_d r_k^d$. Summing over non-overlapping regions yields the lower bound. The dimension effect follows from the known formula for ω_d. □

3.2 Freidlin–Wentzell Large Deviations

Theorem 3.3 (Freidlin–Wentzell Estimate for Basin Transitions). *In the small-noise limit $D_0 \to 0$, the expected first passage time from the basin of attractor $\mathbf{x}_a^*$ to another attractor $\mathbf{x}_b^*$ satisfies:*

$$\mathbb{E}[\tau_{a\to b}] \asymp \exp\left(\frac{\Delta V_{a\to b}}{D_0}\right), \tag{9}$$

where $\Delta V_{a\to b}$ is the minimum barrier height (quasi-potential) from $\mathbf{x}_a^*$ to the basin boundary:

$$\Delta V_{a\to b} = \inf_{\gamma:\mathbf{x}_a^* \to \partial\mathcal{B}(\mathbf{x}_a^*)} \sup_{t\in[0,1]} V(\gamma(t)) - V(\mathbf{x}_a^*). \tag{10}$$

This estimate elevates the Lyapunov function to a ***quasi-potential*** *— the rate of noise-driven civilization phase transitions is governed by the minimum potential barrier between basins.*

Proof. Freidlin–Wentzell large deviation theory formalizes the trajectory probability of SDE (**??**) as a path integral. In the limit $D_0 \to 0$, trajectory probability is dominated by the action functional $S_T[\mathbf{x}] = \frac{1}{4D_0}\int_0^T \|\dot{\mathbf{x}}_t + \nabla V(\mathbf{x}_t)\|^2\,dt$. The minimum-action path is the most likely transition path connecting two attractors, with action equal to the quasi-potential barrier height. The logarithmic equivalence of the first passage time follows from Varadhan's lemma. □

3.3 Basin Stability Phase Diagram

Definition 3.4 (Stability Phase Diagram). *In the institutional parameter space (x_1, x_2, x_3) (C5's triple-lever space), define:*

- $\lambda > 0$ ***region (stable basin):*** $\mathcal{R}_{stab} = \{(x_1, x_2, x_3) : \lambda_{\min}(\nabla^2 V) > 0\}$

- $\lambda = 0$ ***critical surface:*** $\mathcal{S}_{crit} = \{(x_1, x_2, x_3) : \lambda_{\min}(\nabla^2 V) = 0\}$
- $\lambda < 0$ ***region (unstable region):*** $\mathcal{R}_{unstab} = \{(x_1, x_2, x_3) : \lambda_{\min}(\nabla^2 V) < 0\}$

The critical surface $\mathcal{S}_{crit}$ is a codimension-1 $(d-1)$-manifold separating the stable basin from the unstable region.

Definition (Stability Phase Diagram). In the institutional parameter space (x_1, x_2, x_3) (C5's triple-lever space), we define the stable, critical, and unstable regions as above. The critical surface $\mathcal{S}_{\text{crit}}$ is a codimension-1 $(d-1)$-manifold separating the stable basin from the unstable region.

Example 3.5 (Toy Phase Diagram for 3 Institutional Dimensions)**.** *Consider a simple Lyapunov function of the form:*

$$V(x_1, x_2, x_3) = (x_1 - a)^2 + (x_2 - b)^2 + (x_3 - c)^2 - \gamma(x_1 x_2 + x_2 x_3 + x_3 x_1), \tag{11}$$

where (a, b, c) is the optimal institutional configuration and $\gamma \geq 0$ is the institutional coupling strength. The Hessian matrix is:

$$\nabla^2 V = \begin{pmatrix} 2 & -\gamma & -\gamma \\ -\gamma & 2 & -\gamma \\ -\gamma & -\gamma & 2 \end{pmatrix}. \tag{12}$$

Eigenvalues: $\mu_1 = 2 + \gamma$ (two-fold degenerate), $\mu_2 = 2 - 2\gamma$. Therefore:

- *When $\gamma < 1$: all eigenvalues > 0, $\lambda = 2 - 2\gamma > 0$ (stable basin)*
- *When $\gamma = 1$: $\lambda = 0$ (critical surface)*
- *When $\gamma > 1$: $\lambda < 0$ (unstable — excessive institutional coupling induces fragility)*

*Physical meaning: when institutional coupling (γ) is too strong, perturbations in one dimension are transmitted to other dimensions through coupling, amplifying rather than decaying — the system loses local stability. This is **coupling-induced instability**.*

Example (Toy Phase Diagram). With the simple Lyapunov form above, the eigenvalues yield the stability condition $\gamma < 1$. Physical meaning: when institutional coupling (γ) is too strong, perturbations in one dimension are transmitted to others through coupling, amplifying rather than decaying — the system loses local stability. This is coupling-induced instability.

4 Minimal Institutional Core (MIC)

4.1 MIC Definition

Definition 4.1 (Minimal Institutional Core)**.** *The **Minimal Institutional Core** $\mathcal{M} \subseteq \mathcal{X}$ is the smallest subset of institutional configurations satisfying:*

(i) ***Stability condition:*** *For all $\mathbf{x} \in \mathcal{M}$, $\lambda(\mathbf{x}) > 0$ (inside the stable basin);*

(ii) ***Reachability condition:*** *For any initial state $\mathbf{x}_0 \in \mathcal{X}$, there exists a finite time $t < \infty$ and a feasible institutional transition path such that $\mathbf{x}_t \in \mathcal{M}$;*

(iii) ***Minimality condition:*** *No proper subset $\mathcal{M}' \subsetneq \mathcal{M}$ satisfies both (i) and (ii).*

Intuitively, MIC is the smallest set of institutional functions that a civilization ***must*** *possess to maintain long-term stability. Any institutional configuration with fewer dimensions than the MIC necessarily yields $\lambda \leq 0$ — the civilization cannot be stable.*

Definition (Minimal Institutional Core). The **Minimal Institutional Core** $\mathcal{M} \subseteq \mathcal{X}$ is the smallest subset of institutional configurations satisfying stability, reachability, and minimality conditions above. Intuitively, MIC is the smallest set of institutional functions that a civilization must possess to maintain long-term stability. Any institutional configuration with fewer dimensions than the MIC necessarily yields $\lambda \leq 0$ — the civilization cannot be stable.

4.2 MIC Dimension Lower Bound Theorem

Theorem 4.2 (MIC Dimension Lower Bound)**.** *Let the civilization institutional phase space $\mathcal{X} \subseteq \mathbb{R}^d$ be equipped with a C^2 Lyapunov function V. If there exists an attractor basin $\mathcal{B}_+ \neq \emptyset$ with $\lambda > 0$, then the institutional dimension d satisfies:*

$$d \geq 3. \tag{13}$$

Equivalently, ***any civilization with fewer than 3 independent institutional dimensions cannot possess a stable attractor with*** *$\lambda > 0$. The minimal feasible MIC dimension is $d_{MIC} = 3$.*

The three minimal institutional dimensions correspond to the attractor versions of C5's triple suppression levers:

D1 Information Management: *A civilization must possess the ability to regulate information flow — neither fully transparent ($x_1 = 0$) nor fully blocked ($x_1 = 1$), but finding a stable equilibrium between the two.*

D2 Force Monopoly & Distribution: *A civilization must concentrate force to defend against external threats while preventing force from being used for internal oppression — requiring a delicate balance of force institutions.*

D3 Legitimacy Production: *A civilization must continuously produce and maintain a belief framework that renders its institutional arrangements accepted as legitimate — requiring the reproduction of a meaning system.*

Proof. Suppose $d < 3$, i.e., $d \in \{1, 2\}$.

Case $d = 1$**:** $\mathcal{X} \subseteq \mathbb{R}$. The Hessian of the Lyapunov function $V(x)$ reduces to the scalar $V''(x)$. At a local minimum $\mathbf{x}^*$, $V''(x^*) > 0$ (strict convexity), so $\lambda = V''(x^*) > 0$. However, any noise $D_0 > 0$ in a one-dimensional system causes the system to cross any finite barrier with probability 1 (recurrence of one-dimensional diffusion). Hence no genuine basin exists — the system visits all states in the long run, with no true stable attractor. A $d = 1$ civilization necessarily collapses under stochastic shocks.

Case $d = 2$**:** Two-dimensional systems can possess stable attractors (e.g., limit cycles or point attractors). However, two-dimensional continuous dynamical systems are constrained by the Poincaré–Bendixson theorem — the long-term behavior can only be: (a) convergence to an equilibrium, (b) convergence to a limit cycle, or (c) divergence to infinity. Two-dimensional systems cannot exhibit **strange attractors** or **complex multi-basin structures**. Furthermore, two-dimensional diffusion is neighborhood-recurrent —

any noise causes the system to leave an arbitrarily small basin in finite time. Therefore a $d = 2$ civilization can only be stable in the zero-noise limit; any realistic noise induces basin transitions.

Conclusion: $d \geq 3$ is necessary for a robust $\lambda > 0$ attractor basin. Three- and higher-dimensional systems can support: (a) multiple separated attractor basins, (b) non-trivial basin boundaries, and (c) metastability under noise. □

Remark 4.3 (Correspondence with C5). *C5's triple suppression mechanisms — information isolation, force monopoly, belief legitimacy — are reinterpreted in C6 as* ***the three independent dimensions of the MIC****. C5 proved: when suppression along these dimensions is sufficient, $\kappa_{eff} \to 0$ and civilization can be long-lived. C6 further proves:* ***the very existence of these three dimensions is necessary for*** *$\lambda > 0$ — not because they suppress κ, but because they constitute the minimal independent basis of institutional phase space, enabling a non-degenerate Lyapunov function.*

4.3 Computable Characterization of MIC

Proposition 4.4 (Hessian Characterization of MIC). *An institutional configuration* $\mathbf{x}$ *belongs to the MIC if and only if:*

(i) $\nabla V(\mathbf{x}) = \mathbf{0}$ *(institutional equilibrium condition);*

(ii) $\lambda_{\min}(\nabla^2 V(\mathbf{x})) > 0$ *(local stability condition);*

(iii) No subset of institutional dimensions $\mathcal{I} \subsetneq \{1, \ldots, d\}$ exists such that the subsystem restricted to $\mathcal{I}$ also satisfies (i) and (ii) (minimality condition).

This provides a computable MIC detection algorithm: search all critical points with positive-definite Hessian in institutional phase space, then verify minimality through dimensional elimination.

Proposition (Hessian Characterization of MIC). An institutional configuration $\mathbf{x}$ belongs to the MIC iff it satisfies the three conditions above. This provides a computable MIC detection algorithm: search all critical points with positive-definite Hessian in institutional phase space, then verify minimality through dimensional elimination.

5 λ–T Relationship and Civilization Lifespan

5.1 First Exit Time and Lyapunov Exponent

Theorem 5.1 (Exponential λ–T Relationship). *Suppose a civilization is currently in the basin $\mathcal{B}(\mathbf{x}^*)$ of attractor $\mathbf{x}^*$, with $\lambda = \lambda_{\min}(\nabla^2 V(\mathbf{x}^*)) > 0$. Then the first exit time τ_{exit} (when the civilization leaves the basin, i.e., institutional collapse) satisfies:*

$$\mathbb{E}[\tau_{exit}] \asymp \exp\left(\frac{\lambda \cdot \rho(\mathbf{x}^*)^2}{2D_0}\right) \quad \text{as } D_0 \to 0, \tag{14}$$

where $\rho(\mathbf{x}^*) = \min_{\mathbf{y} \in \partial\mathcal{B}(\mathbf{x}^*)} \|\mathbf{y} - \mathbf{x}^*\|$ is the basin radius (shortest distance from the attractor to the basin boundary).

Let $\tau_{char} = \rho(\mathbf{x}^*)^2/(2D_0)$ be the characteristic diffusion time, then:

$$T \equiv \mathbb{E}[\tau_{exit}] \propto \exp\left(\lambda \cdot \tau_{char}\right). \tag{15}$$

Core implication: *Civilization expected lifespan T depends exponentially on Lyapunov exponent λ — a small increase in λ (through institutional improvement) yields a large increase in T.*

Proof. From Freidlin–Wentzell theory (Theorem **??**), the log-asymptotic of the first exit time equals the quasi-potential barrier divided by noise intensity. Near the attractor, V is well-approximated by its quadratic expansion:

$$V(\mathbf{x}) \approx V(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T \nabla^2 V(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*).$$

The minimum V value on the basin boundary $\partial\mathcal{B}(\mathbf{x}^*)$ occurs at the point closest to $\mathbf{x}^*$, and:

$$\min_{\mathbf{y}\in\partial\mathcal{B}} V(\mathbf{y}) - V(\mathbf{x}^*) \approx \frac{1}{2}\lambda_{\min}(\nabla^2 V) \cdot \rho(\mathbf{x}^*)^2 = \frac{1}{2}\lambda \cdot \rho(\mathbf{x}^*)^2.$$

Substituting into Theorem **??** yields (**??**). □

5.2 Lambda Composition and Institutional Investment

Proposition 5.2 (Additive Decomposition of λ)**.** *Suppose the Hessian of the Lyapunov function decomposes across MIC dimensions as:*

$$\nabla^2 V = \sum_{k=1}^{d} \mathbf{H}_k, \tag{16}$$

where $\mathbf{H}_k$ is the contribution of the k-th institutional dimension. Then:

$$\lambda = \lambda_{\min}\left(\sum_{k=1}^{d} \mathbf{H}_k\right) \geq \sum_{k=1}^{d} \lambda_{\min}(\mathbf{H}_k), \tag{17}$$

where the inequality follows from Weyl's eigenvalue inequality (superadditivity of the minimum eigenvalue).

Institutional investment interpretation: *Investment I_k in each institutional dimension k produces a Hessian contribution $\mathbf{H}_k(I_k)$. The total stability index λ is the (approximately) linear superposition of contributions from each dimension. This implies:*

- ***Increasing returns:*** *As institutional dimensions increase, λ grows at a superlinear rate (superadditivity of eigenvalues);*
- ***Diminishing marginal returns:*** *Over-investment in a single dimension ($\mathbf{H}_k$ saturation) yields diminishing marginal contributions to λ;*
- ***Diversity advantage:*** *Moderate investment across multiple dimensions boosts λ more effectively than concentrated investment in a single dimension.*

Proposition (Additive Decomposition of λ). When the Hessian decomposes additively across MIC dimensions, the stability index satisfies the superadditivity bound above. Interpretation: diversified institutional investment across multiple dimensions is more effective at boosting λ than concentrated investment in a single dimension.

6 Phase Transitions and Critical Phenomena

6.1 Critical Surface in Institutional Parameter Space

Theorem 6.1 (Stability Phase Transition Theorem). *Let $V(\mathbf{x};\boldsymbol{\theta})$ be a family of Lyapunov functions depending on institutional parameters $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p$. Then there exists a critical hypersurface $\Theta_{crit} \subset \Theta$ such that:*

- *When $\boldsymbol{\theta} \in \Theta_+$ (one side of Θ_{crit}), $\lambda(\boldsymbol{\theta}) > 0$ — stable phase;*
- *When $\boldsymbol{\theta} \in \Theta_-$ (the other side), $\lambda(\boldsymbol{\theta}) < 0$ — collapse phase;*
- *On Θ_{crit}, $\lambda(\boldsymbol{\theta}) = 0$ — critical phase, exhibiting **critical slowing down** and **fluctuation amplification**.*

The critical surface Θ_{crit} is defined by the equation $\det(\nabla^2_{\mathbf{x}} V(\mathbf{x}^*(\boldsymbol{\theta});\boldsymbol{\theta})) = 0$, where $\mathbf{x}^*(\boldsymbol{\theta})$ is the parameter-dependent attractor location.

Proof. By the implicit function theorem, the attractor location $\mathbf{x}^*(\boldsymbol{\theta})$ satisfies $\nabla_{\mathbf{x}} V(\mathbf{x}^*(\boldsymbol{\theta});\boldsymbol{\theta}) = \mathbf{0}$. Differentiating with respect to $\boldsymbol{\theta}$ yields $\nabla^2_{\mathbf{x}} V \cdot \frac{\partial \mathbf{x}^*}{\partial \boldsymbol{\theta}} + \nabla^2_{\mathbf{x}\boldsymbol{\theta}} V = \mathbf{0}$. When $\det(\nabla^2_{\mathbf{x}} V) = 0$, $\frac{\partial \mathbf{x}^*}{\partial \boldsymbol{\theta}}$ diverges — the sensitivity of attractor location to parameters diverges, signaling a phase transition. □

6.2 Critical Exponents and Universality

Conjecture 6.2 (Universality Class of Civilization Stability Phase Transitions). *Along a generic transverse direction in institutional parameter space, the scaling behavior of λ near the critical surface satisfies:*

$$\lambda(\theta) \sim \begin{cases} C_+ \cdot (\theta - \theta_c)^\beta, & \theta > \theta_c \text{ (stable phase)}, \\ 0, & \theta = \theta_c \text{ (critical point)}, \\ -C_- \cdot (\theta_c - \theta)^\beta, & \theta < \theta_c \text{ (collapse phase)}, \end{cases} \tag{18}$$

where θ is the parameter projection along the most unstable direction (the eigenvector corresponding to the smallest eigenvalue of $\nabla^2 V$), and θ_c is the critical value.

Based on Landau-type mean-field arguments, we expect $\beta = 1/2$ (mean-field exponent). For institutional dimension $d = 3$, fluctuation corrections may shift β away from the mean-field value — this is an open problem.

Conjecture (Universality Class). Near the critical surface, λ exhibits power-law scaling with exponent β. Mean-field arguments give $\beta = 1/2$. For $d = 3$, fluctuation corrections may shift β away from the mean-field value — this is an open problem.

6.3 Noise-Induced Phase Transitions

Theorem 6.3 (Noise-Induced Basin Collapse). *There exists a critical noise intensity $D_c > 0$ such that when $D_0 > D_c$, the probability of the system remaining in the basin for any finite time is less than an arbitrarily specified threshold $\epsilon > 0$. Specifically:*

$$D_c = \frac{\lambda \cdot \rho(\mathbf{x}^*)^2}{2\log(1/\epsilon)}. \tag{19}$$

*When $D_0 > D_c$, even though $\lambda > 0$ (deterministically stable), the noise intensity is sufficient to trigger frequent basin transitions on observational time scales — the civilization is no longer stable in a **statistical sense**.*

Proof. From (??), $\mathbb{E}[\tau_{\text{exit}}] \asymp \exp(\lambda\rho^2/(2D_0))$. Let the observational time scale be T_{obs}. Require $\mathbb{P}(\tau_{\text{exit}} > T_{\text{obs}}) > 1 - \epsilon$. By Markov's inequality, $\mathbb{P}(\tau_{\text{exit}} \leq T_{\text{obs}}) \leq T_{\text{obs}}/\mathbb{E}[\tau_{\text{exit}}]$. Setting this upper bound $< \epsilon$ and solving yields $D_0 < \lambda\rho^2/(2\log(T_{\text{obs}}/\epsilon))$. Taking the most conservative case $T_{\text{obs}} = 1$ gives (??). □

7 Multi-Institution Dynamics and Competitive MIC

7.1 N-Institution Lotka–Volterra Model

Definition 7.1 (N-Institution Competitive Dynamics). *Consider N institutional entities, where each institution i is parameterized by its stability index $\lambda_i \in \mathbb{R}_{\geq 0}$. The evolution of institution i follows a generalized Lotka–Volterra SDE:*

$$d\lambda_i(t) = \lambda_i(t) \left[r_i - \sum_{j=1}^{N} A_{ij}\lambda_j(t) \right] dt + \sigma_i \, dW_i(t), \tag{20}$$

where:

- $r_i > 0$*: intrinsic stability growth rate of institution i (institutional efficiency);*
- $A_{ij} \geq 0$*: inter-institution competition matrix, where A_{ij} represents the competitive pressure of institution j on institution i;*
- $\sigma_i \geq 0$*: environmental noise intensity faced by institution i;*
- $W_i(t)$*: independent Brownian motions.*

The biological/ecological analogy: each institution i is like a species in an ecosystem, with λ_i as its population size, r_i as its intrinsic growth rate, and A_{ij} as the interspecies competition coefficient.

Definition (N-Institution Competitive Dynamics). N institutional entities coevolve under generalized Lotka–Volterra SDE dynamics, where each institution's stability index λ_i functions as its population size and the competition matrix A_{ij} encodes institutional rivalry.

7.2 Emergence of Competitive MIC

Theorem 7.2 (Competitive MIC Existence Theorem). *Consider the N-institution system (??). If the following conditions hold:*

(i) ***Feasibility condition:*** *There exists $\lambda^* \in \mathbb{R}^N_{>0}$ such that $A\lambda^* = \mathbf{r}$ (positive interior equilibrium exists);*

(ii) ***Stability condition:*** *All leading principal minors of the competition matrix A are positive (A is a P-matrix);*

(iii) ***Diagonal dominance:*** *$A_{ii} > \sum_{j\neq i} A_{ij}$ (self-competition dominates cross-institutional competition).*

Then: (a) the system has a unique globally stable interior equilibrium λ^*; (b) the civilization Lyapunov exponent at equilibrium is $\lambda_{total} = \sum_i w_i\lambda_i^*$, where $w_i > 0$ are institution importance weights; (c) the set of surviving institutions constitutes the civilization's ***realized MIC****.*

When N is sufficiently large and the competition matrix satisfies the above conditions, the system's realized MIC converges to the theoretical MIC (Definition **??***).*

Proof. Conditions (i)–(ii) guarantee the existence and uniqueness of the interior equilibrium $\lambda^* = A^{-1}\mathbf{r} > \mathbf{0}$. Define the Lyapunov function:

$$V(\boldsymbol{\lambda}) = \sum_{i=1}^{N} w_i \left[\lambda_i - \lambda_i^* - \lambda_i^* \log\left(\frac{\lambda_i}{\lambda_i^*}\right)\right],$$

where $w_i > 0$ are weights to be determined. The time derivative along the deterministic flow is:

$$\frac{d}{dt}V = -\frac{1}{2}\sum_{i,j}(\lambda_i - \lambda_i^*)(w_iA_{ij} + w_jA_{ji})(\lambda_j - \lambda_j^*) \leq 0,$$

which is strictly negative definite when A satisfies condition (iii). Therefore $\boldsymbol{\lambda}^*$ is a global attractor. The positivity of λ_{total} follows from the positivity of each component. □

7.3 Institutional Competition Phase Diagram

Table 1: Dynamical Phases of N-Institution Systems

Phase	$\det(A)$	**Equilibrium Type**	**Civilization Meaning**
Coexistence	> 0 (P-matrix)	Positive interior equilibrium $\lambda_i^* > 0\ \forall i$	Multi-institution syne tic stability, complete M
Partial Exclusion	≥ 0 (degenerate)	Some $\lambda_i^* = 0$	Some institutional funct lost, MIC deficient
Full Monopoly	indefinite	Only one $\lambda_i^* > 0$	Single institution d nance, high fragility
Total Collapse	—	All $\lambda_i \to 0$	System collapse, no st attractor

Table: Dynamical Phases of N-Institution Systems. The four phases span from full coexistence (complete MIC) to total collapse (no stable attractor).

8 Connection to Existing SCX Theorems

8.1 Unification with Thm12 (C5)

Theorem 8.1 (Thm12–C6 Unification Theorem)**.** *C5's Thm12′ (Coupling-Modified Civilization Collapse Theorem) and C6's λ–T relationship* (**??**) *are equivalent under the following correspondence:*

$$\lambda \longleftrightarrow -\log(\kappa_{\mathit{eff}} \cdot \Delta^2). \tag{21}$$

*That is, C5's coupling suppression mechanism and C6's attractor stability are two sides of the same coin. $\lambda > 0$ is equivalent to $\kappa_{\mathit{eff}} \cdot \Delta^2 < 1$ (effective disruption rate less than 1). C6's Lyapunov perspective provides the **dynamical foundation**, while C5's coupling perspective provides the **mechanistic explanation**.*

Proof. From Thm12′ (C5), $\mathbb{E}[T] = 1/(\alpha \cdot \kappa_{\text{eff}} \cdot \Delta^2)$. From C6's (**??**), $T \propto \exp(\lambda \cdot \tau_{\text{char}})$. When $\kappa_{\text{eff}} \cdot \Delta^2 \ll 1$, taking logarithms: $\log T \approx -\log(\kappa_{\text{eff}} \cdot \Delta^2) - \log \alpha$. Comparing with $\log T \approx \lambda \cdot \tau_{\text{char}} + \text{const}$ yields the correspondence (**??**) (up to constant factors). $\square$

8.2 Connection to Recursive Audit (C4)

Proposition 8.2 (Audit Depth–Lyapunov Exponent Duality). *C4's recursive audit noise decay factor α (the α in Model B) and C6's Lyapunov exponent λ satisfy the duality relation:*

$$\lambda = -\log \alpha. \tag{22}$$

That is, the convergence rate of audit noise in recursive auditing ($\alpha < 1$) is dual to the stability strength of the attractor in civilization dynamics ($\lambda > 0$). Fast-decaying audit noise ($\alpha \ll 1$) is dual to a deeply stable civilization attractor ($\lambda \gg 0$); non-decaying audit noise ($\alpha \geq 1$) is dual to critical or collapsing civilization ($\lambda \leq 0$).

Proposition (Audit Depth–Lyapunov Exponent Duality). C4's recursive audit noise decay factor α and C6's Lyapunov exponent λ satisfy the duality relation $\lambda = -\log \alpha$. Fast-decaying audit noise ($\alpha \ll 1$) is dual to a deeply stable civilization attractor ($\lambda \gg 0$); non-decaying audit noise ($\alpha \geq 1$) is dual to critical or collapsing civilization ($\lambda \leq 0$).

9 Numerical Verification Framework

9.1 Verification Objectives

We numerically verify the following core predictions of C6 through simulation:

V1 MIC dimension hypothesis: $d \geq 3$ is necessary for the existence of a $\lambda > 0$ basin. Simulate systems with $d = 1, 2, 3, 4, 5$ dimensions and verify that systems with $d < 3$ cannot maintain a stable basin under noise.

V2 Lyapunov function convergence: Along gradient flow $\dot{\mathbf{x}} = -\nabla V$, $V(\mathbf{x}_t)$ decreases monotonically and converges to the minimum.

V3 λ–T exponential relationship: Simulate first exit times for different λ values and verify $T \propto \exp(\lambda \cdot \tau_{\text{char}})$.

V4 Phase transition critical behavior: As the parameter γ (institutional coupling strength) crosses the critical value $\gamma_c = 1$, verify the continuous/discontinuous change in λ.

V5 N-institution competitive dynamics: Simulate 2–5 institution systems and verify the conditions for coexistence, partial exclusion, and full monopoly phases.

9.2 Verification Script Structure

The verification script `verify_lambda_attractor.py` contains the following modules:

- **Module 1:** Lyapunov function construction and Hessian computation
- **Module 2:** Deterministic gradient flow simulation (Euler method)
- **Module 3:** SDE trajectory simulation (Euler–Maruyama method)
- **Module 4:** Basin structure and first exit time estimation
- **Module 5:** N-institution Lotka–Volterra dynamics
- **Module 6:** MIC detection and verification
- **Module 7:** Comprehensive test runner

10 Discussion and Outlook

10.1 Theoretical Significance

C6 elevates civilization stability research from C5's mechanism explanation layer to the dynamical foundation layer. Through Lyapunov function and attractor basin theory, we obtain the following profound insights:

(1) **Necessity of institutional dimensions:** The three-dimensional MIC is not merely an empirical observation (C5's triple suppression), but a topological necessity — systems with fewer than three dimensions cannot maintain separate attractor basins under noise.

(2) **Stability–fragility duality:** The same institutional coupling γ can enhance stability when moderate (through inter-dimension synergy) or induce collapse when excessive (coupling-induced instability).

(3) **Exponential sensitivity:** The exponential dependence of civilization lifespan on the Lyapunov exponent implies a **leverage effect of institutional fine-tuning** — small institutional improvements can yield order-of-magnitude lifespan gains.

(4) **Dual role of noise:** Noise is both a destroyer (triggering basin transitions) and an explorer (enabling civilizations to search for better configurations in institutional space) — moderate noise may be optimal.

10.2 Open Problems

C6 opens the following open problems:

OP1 Lyapunov function calibration from real civilization data: How can we infer the specific functional form of $V(\mathbf{x})$ from historical data? What institutional metrics are needed?

OP2 Time-varying MIC: Does the MIC evolve with technological progress? Is the MIC of modern industrial civilizations still $d = 3$, or does it require additional dimensions?

OP3 Quantum extension: Civilization quantum tunneling — can basin boundaries be crossed instantaneously through non-classical paths (e.g., revolutions, black swan events)?

OP4 Networked MIC: When multiple civilizations are coupled through trade, diplomacy, and information networks, how is the joint Lyapunov function defined? Does a global attractor of the civilization network exist?

OP5 Empirical measurement of λ: How can λ be estimated from contemporary institutional indicators? Which civilizations are closest to the $\lambda = 0$ critical boundary?

11 Conclusion

This paper establishes the C6 extension of the SCX Equality Theory framework — the civilization λ attractor design theory. Core contributions include: (1) formal definition of the Lyapunov civilization function $V(\mathbf{x})$, mapping civilization stability to a scalar field on institutional phase space; (2) geometric and topological analysis of the $\lambda > 0$ attractor basin; (3) proof of the existence of the Minimal Institutional Core (MIC) with dimension lower bound $d \geq 3$; (4) Freidlin–Wentzell derivation of the exponential λ–T relationship; (5) Lotka–Volterra model of N-institution competitive dynamics and conditions for competitive MIC emergence; (6) unification and duality with C5 (Thm12) and C4 (recursive audit).

The core message of C6 is: **the long-term stability of civilizations is not a historical accident, but a necessity of the Lyapunov function in institutional phase space — if and only if a civilization establishes a Minimal Institutional Core satisfying $\lambda > 0$, its dynamics are irreversibly attracted to a stable basin.** Civilizations lacking the MIC, however prosperous they may appear on the surface, reside in the $\lambda < 0$ repeller region — collapse is only a matter of time.

A Appendix A: Mathematical Details

A.1 Euler–Maruyama Discretization

The Euler–Maruyama discretization scheme for SDE (??) with time step Δt is:

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \nabla V(\mathbf{x}_n) \cdot \Delta t + \sqrt{2D_0 \Delta t} \cdot \boldsymbol{\xi}_n, \tag{23}$$

where $\boldsymbol{\xi}_n \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ is the standard d-dimensional Gaussian vector. The strong convergence order is $\mathcal{O}(\Delta t^{1/2})$.

A.2 Krasnoselskii Existence for Lotka–Volterra Equilibrium

The existence of the interior equilibrium $\lambda^* > \mathbf{0}$ in Theorem **??** is guaranteed by the Krasnoselskii fixed-point theorem. Define the operator $\mathcal{T} : \mathbb{R}^N_{\geq 0} \to \mathbb{R}^N_{\geq 0}$ as $\mathcal{T}(\lambda) = D(\lambda)^{-1}\mathbf{r}$, where $D(\lambda) = \mathrm{diag}(\sum_j A_{1j}\lambda_j, \ldots, \sum_j A_{Nj}\lambda_j)$. A continuous self-map on a compact convex set has a fixed point.

A.3 Variance Divergence at Critical Slowing Down

Near the critical surface Θ_{crit}, $\lambda \to 0$. The variance of the stationary distribution (??) is:

$$\mathrm{Var}_\pi[\mathbf{x}] \approx \frac{D_0}{\lambda} \quad \text{as } \lambda \to 0^+. \tag{24}$$

That is, as civilization approaches the critical state, the variance of institutional configuration diverges as $1/\lambda$ — this **critical fluctuation amplification** provides an early warning signal for the impending phase transition.

B Appendix B: N-Institution Simulation Parameters

Table 2: Numerical Simulation Parameters

Parameter	**Symbol**	**Default**
Institutional dimension	d	3
Time step	Δt	0.01
Simulation duration	T_{sim}	100.0
Noise intensity	D_0	0.01
Number of institutions	N	3
Intrinsic growth rate	r_i	$\sim \mathcal{U}(0.5, 1.5)$
Competition matrix	A	$A_{ii} = 1$, $A_{ij} \sim \mathcal{U}(0.1, 0.5)$
Coupling strength	γ	0.5 (stable phase)

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References

[1] SCX Gauge Theory Working Group. SCX Gauge Theory: Gauge Equivalence in Audit Data Processing and Cercis Space Structure. Technical Report C1–C6, 2026.

[2] SCX Equality Theory Research Group. Civilization Inequality Gauge: κ Suppression and Desuppression (C5 Extension). Technical Report, 2026.

[3] SCX Equality Theory Research Group. Consciousness Audit Boundary: Noise Divergence and Compactness of Recursive Self-Audit (C4 Extension). Technical Report, 2026.

[4] M.I. Freidlin and A.D. Wentzell. *Random Perturbations of Dynamical Systems*. Springer, 3rd edition, 2012.

[5] A.M. Lyapunov. *The General Problem of the Stability of Motion*. Taylor & Francis, 1992 (original 1892).

[6] S.R.S. Varadhan. *Large Deviations and Applications*. SIAM, 1984.

[7] C.W. Gardiner. *Handbook of Stochastic Methods*. Springer, 4th edition, 2009.

[8] J.D. Murray. *Mathematical Biology I: An Introduction*. Springer, 3rd edition, 2002.

[9] M. Scheffer et al. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009.

[10] P. Turchin. *Historical Dynamics: Why States Rise and Fall*. Princeton University Press, 2003.

[11] J. Diamond. *Collapse: How Societies Choose to Fail or Succeed*. Viking, 2005.

Verification Script

This paper is accompanied by a complete numerical verification script `verify_lambda_attractor.py` that can be run independently to reproduce all numerical results. Run as follows:

```
cd papers/scx_lambda_attractor/
python verify_lambda_attractor.py
```

The script contains the following verification modules:

- V1: MIC dimension verification ($d = 1, 2, 3, 4, 5$ comparison)
- V2: Lyapunov function convergence (gradient flow monotonic decrease)
- V3: λ–T exponential relationship (Freidlin–Wentzell fitting)
- V4: Phase transition critical behavior (γ scan and λ scaling)
- V5: N-institution competitive dynamics (coexistence/exclusion/monopoly phase diagram)
- V6: Noise robustness (basin residence time under different D_0)